Tensor computations

The four-exception conjecture for alternating tensor border-rank varieties

Difficulty: extreme **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: Excluding all additional defective Grassmannian secants is a longstanding classification barrier despite extensive proved parameter ranges. Its direct importance is to specialists in alternating tensor decompositions and secant geometry.

Problem statement

For integers $p \geq 3$ and $N \geq 2p$, let

$$C_{p,N} = \{v_1 \wedge \cdots \wedge v_p : v_1, \ldots, v_p \in \mathbb{C}^N\} \subset \bigwedge^p \mathbb{C}^N.$$

For every integer $s \geq 1$, let $Y_s = \overline{\{x_1 + \cdots + x_s : x_i \in C_{p,N}\}}^{\mathrm{Zar}}$. Its expected affine dimension is

$$E_{p,N,s} = \min\left\{\binom{N}{p},\ s(p(N-p)+1)\right\}.$$

Is $\dim Y_s = E_{p,N,s}$ except precisely for the following four triples, with the indicated actual affine dimensions?

(p, N, s)	$\dim Y_s$
$(3, 7, 3)$	34
$(4, 8, 3)$	50
$(4, 8, 4)$	64
$(3, 9, 4)$	74

Equivalently, the listed cases should exhaust the defective secant varieties of Grassmannians after using duality to impose $N \geq 2p$. The restriction $p \geq 3$ excludes the separate, already understood skew-symmetric matrix case. The dimension formula uses the affine cone, hence the $+1$ in each summand's parameter count.

Why it matters

This predicts the dimensions of spaces of alternating tensors that admit a given number of decomposable wedge terms, including limiting decompositions. It gives the alternating-tensor analogue of generic rank formulas for symmetric tensors and identifies the exceptional formats where parameter counting mispredicts approximation complexity.

References and status check

1. K. Baur, J. Draisma, and W. A. de Graaf, *Secant dimensions of minimal orbits: computations and conjectures*, Experimental Mathematics **16** (2007), 239–250, Conjecture 4.1; the original statement uses $p > 2$ and $2p \leq N$.
2. A. Oneto and E. Ventura, *Ranks of tensors: geometry and applications*, 2025 survey, §2.2.3, Conjecture 2.16; its projective notation has $(k, n) = (p-1, N-1)$.
3. A. Taveira Blomenhofer and A. Casarotti, *Nondefectivity of invariant secant varieties*, Full preprint, v2, §4, Theorem 4.3 and Remark 4.4.

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Checked the original-convention exception list against the 2025 restatement and Blomenhofer–Casarotti v2, Theorem 4.3 and Remark 4.4, with targeted later Grassmannian-defectivity searches. Theorem 4.3 proves nondefectivity for substantive general rank ranges, so Partially resolved applies to the displayed all-parameter target. These bounds leave intermediate ranks and do not exhaust the possible exceptions. The 2025 survey still states the four-exception conjecture; no full classification was located. The listed dimensions retain the affine-cone convention.